Tightened Readout Bounds, Optimal λ_2, and Joint Parameter Optimization

[Supplementary Technical Note]

I BACKGROUND & PROBLEM STATEMENT

The constructive realization document bounds classification error by

$$\text{error} \leq k \exp\left(-\frac{\gamma^2}{8\sigma_r^2}\right), \quad \sigma = \rho_0 \sigma_{EEG} \|B^T\|_2$$

treating noise as isotropic. However, here I tighten these bounds, considering within-class anisotropy ($\sigma_r^2 \to \sigma_r^2/r^2$) and derive λ_2^* maximizing classification accuracy. Additionally, I determine the joint optimal parameter vector $\Theta^* = (r_0^*, \delta^*, K_{PLL}^*, \rho_0^*)$ minimizing the expected proportional tracking error.

$$\mathcal{E}(\Theta) := \mathbb{E}\left[\|\hat{s}(t) - s^{(i)}(t)\|_2^2\right]$$

subject to the following conditions:

1) $\rho_{sep} > f(r_0, \alpha)$
2) $K_{PLL} < \dfrac{\gamma(r_0, \delta)}{2\|\xi_i\|_2}$
3) $\rho_0 > \rho^{min}(r_0, \delta, \alpha)$
4) $r_0 \in (0,1)$, $\delta < r_0$, $K_{PLL} > 0$, $\rho_0 > 0$

Additionally, we accept the following assumptions:

1) Each reservoir attractor $\Lambda_i \subseteq S^1$ is a normally hyperbolic limit cycle with gap $\gamma > 0$. The within-class covariance decomposes as $\Sigma_w^{(i)} = \sigma_{tang}^2 P_{tang}^{(i)} + \sigma_{norm}^2 P_{norm}^{(i)}$, where $P_{tang}^{(i)}$, $P_{norm}^{(i)}$ project onto the tangential/normal subspaces, and $\sigma_{tang}^2 \gg \sigma_{norm}^2$

2) Reservoir noise is isotropic with $\sigma_r = \rho_0 \sigma_{EEG} \|B^T\|_2$. Normal hyperbolicity suppresses transverse noise: $\sigma_{norm} = \sigma_r / r$

3) $C \in \mathbb{R}^{k \times N}$ is the ridge regression readout. The discriminant direction for pair (j, ℓ) is $c_{j\ell} = (C e_j - C e_\ell)/\|C e_j - C e_\ell\|_2$

4) $d_H(A_j, A_\ell) > \eta$ for all $j \neq \ell$, guaranteed by $\lambda_2 \in (\underline{\lambda}_2, \bar{\lambda}_2)$

II TIGHTENED BOUND VIA ANISOTROPY

Lemma 2.1 (Transverse noise) The within class readout variance in direction $c_{j\ell}$ satisfies

$$\mathrm{Var}_{r \sim A_j}\left[c_{j\ell}^T C r\right] \leq \sigma_{\text{long}}^2 \cos^2\phi_{j\ell} + \sigma_{\text{norm}}^2 \sin^2\phi_{j\ell}$$

where $\phi_{j\ell}$ is the angle between $\mu_j - \mu_\ell$ and the tangent space of A_j. For attractors in distinct frequency bands, $\phi_{j\ell} \approx \pi/2$ (inter-attractor direction is transverse to limit cycles). Thus

$$\mathrm{Var}_{r \sim A_j}\left[c_{j\ell}^T C r\right] \approx \sigma_{\text{norm}}^2 = \frac{\sigma_r^2}{r^2}$$

The squared Mahalanobis distance between attractor means μ_j and μ_ℓ is then

$$\Delta_{j\ell}^2 \approx \frac{\eta_{j\ell}^2 r^2}{\sigma_r^2}.$$

where $\eta_{j\ell} = \|P_{\text{norm}}(\mu_j - \mu_\ell)\|_2 > \eta$. The minimum Fisher discriminant ratio is

$$F_{\min} = \min_{j \neq \ell} \frac{(\mu_j - \mu_\ell)^T \Sigma_W^{-1} (\mu_j - \mu_\ell)}{2} \approx \frac{\eta^2 r^2}{2\sigma_r^2}$$

Theorem 2.2 (Tightened Accuracy Bound) Under the assumptions I previously made, the classification error satisfies

$$\text{error} \leq k^2 \exp\left(-\frac{\eta^2 r^2}{8\sigma_r^2}\right) = k^2 \exp\left(-\frac{\eta^2 r^2}{8\rho_C^2 \sigma_{\text{eff}}^2 \|B^T\|_2^2}\right)$$

This improves the original bound by replacing isotropic σ_r^2 with effective noise $\sigma_{\text{norm}}^2 = \sigma_{\text{eff}}^2 / r^2$, a factor r^2 gain in the exponent.

Proof. By ~~the Gaussian tail bound~~ and Lemma 2.1 with $\eta_{i,\ell} \geq \eta$:

~~$$P(\text{error}_{i,\ell}) \leq \exp\left(-\frac{\eta_{i,\ell}^2}{\quad}\right)$$~~

For a k-class classifier, the probability of misclassifying class ℓ as class ℓ is bounded by the tail probability of the projected noise exceeding half the inter-class distance in the readout space. By gaussian tail bound $P(z > t) \leq \exp(-t^2/2)$ for $z \sim \mathcal{N}(0,1)$:

$$P(\text{error}_{j,\ell}) \leq \exp\left(\frac{\|\mu_j - \mu_\ell\|^2}{8\,\mathrm{Var}[c_{j,\ell}^T c_r]}\right) \leq \exp\left(\frac{-\eta_{j,\ell}^2}{8\,\sigma_{norm}^2}\right)$$

$$= \exp\left(-\frac{\eta_{j,\ell}^2 r^2}{8\,\sigma_r^2}\right)$$

The first inequality uses the Gaussian tail; the second uses Lemma 2.1 with $\eta_{j,\ell} > \eta$ and the variance approximation. Union bounding over all $\binom{k}{2} \leq k^2/2$ pairs and bounding $k^2/2 \leq k^2$:

$$P(\text{error}) < \sum_{j \neq \ell} P(\text{error}_{j,\ell}) \leq k^2 \exp\left(-\frac{\eta^2 r^2}{8\sigma_r^2}\right)$$

Substituting $\sigma_r = \rho_0 \sigma_{EEA} \|B^T\|_2$, the bound becomes $k^2 \exp(-\eta^2 r^2 / (8\rho_0^2 \sigma_{EEA}^2 \|B^T\|_2^2))$. This exposes a trade off: larger ρ_0 activates limit cycles (Neimark-Sacker condition) but amplifies noise and degrades the bound.

III JOINT PARAMETER OPTIMIZATION

Error Decomposition

I begin by making the three-term decomposition explicit and lying terms to design parameters.

Proposition 3.2 The explicit proportional tracking error satisfies

$$\mathcal{E}(\theta) \le \|C_{reg}^{(i)}\|_2^2 \left(\rho_0^2 \sigma_{EEG}^2 \|B^T\|_2^2 + \frac{c_1^2 \rho_0^2 \sigma_{EEG}^2}{(r_0 - \delta)^2} + A_j^2 \cdot \frac{\sigma_\phi^2}{\alpha K_{PLL} \|\xi_j\|_2^2 (2 - \alpha K_{PLL} \|\xi_j\|_2^2)}\right),$$

where c_1 is a constant depending on the ESP kernel and the spectral gap $\Delta = r_0 - \delta$. Here, term 1 is the eeg noise floor, term 2 is the transverse attractor error, term 3 is the phase tracking error. □

~~Proof By expanding the e~~

~~Parameter Dependencies~~

Constrained Nonlinear Programme and KKT Analysis

~~The key structural observation motivating the optimization is stated here:~~

(~~Why Sequential Optimization Fails~~) Let $\theta = (r_0, \delta, K_{PLL}, \rho_0) \in \in \mathbb{R}^4$ and define

$$\min_{\theta} \mathcal{E}(\theta) = \|C_{reg}^{(i)}\|_2^2 \left(\rho_0^2 C_n + \frac{c_1^2 \rho_0^2 \sigma_{EEG}^2}{(r_0 - \delta)^2} + \frac{A_j^2 \sigma_\phi^2}{\alpha K_{PLL} \|\xi_j\|_2^2 (2 - \alpha K_{PLL} \|\xi_j\|_2^2)}\right)$$

subject to the original conditions,
where $C_n = \sigma_{EEG}^2 \|B^T\|_2^2$

Theorem 3.5 (Existence of Optimum) ~~Let $\theta = (r_0, \delta, K_{PLL}, \rho_0) \in \mathbb{R}^4$~~
Under the following regularity conditions:

1) $f(r_0, \alpha)$ is continuous and strictly decreasing in r_0
2) $\rho_0^{min}(r_0, \delta, \alpha)$ is continuous and strictly decreasing in r_0
3) $r(r_0, \delta)$ is continuous and equals $r_0 - \delta$ to leading order

The feasible set $\Theta = \{\theta : C1 \to C4 \text{ hold}\}$ is nonempty and the infimum of $\mathcal{E}$ over Θ is attained at a boundary point of Θ where at least one of C1, C2, C3 is active.

Proof (non-emptiness) For r_0 close to 1, constraint C1 is satisfied (large ρ_{sep}). Choosing δ small makes $r \approx r_0 - \delta$ large, satisfying C2. The activation threshold ρ_0^{min} can then be met with moderate ρ_0, satisfying C3.

Attainment at Boundary: Each of $\mathcal{E}_1, \mathcal{E}_2$ is strictly decreasing as $\rho_0 \to 0$ and $\Delta \to \infty$. The constraint C3 provides a lower bound on ρ_0 that increases as Δ decreases, so the objective has no interior critical point. Hence, the infimum is attained on $\partial\Theta$, where the C1–C3 holds with equality. ▨

Corollary 3.6 (Binding constraint) The identity of the active constraint at the optimum diagnoses the system bottleneck:

- C1 active: insufficient Riemannian separability; improve electrode placement or increase covariance order N_w
- C2 active: phase noise dominates; narrow the bandpass filter to reduce σ_ϕ^2
- C3 active: reservoir is marginally driven; increase N or improve CCA alignment

KKT Conditions

The Lagrangian for the joint optimization problem is

$$\mathcal{L}(\theta, \lambda) = \mathcal{E}(\theta) - \lambda_1 (\rho_{sep} - f(\rho_0, \Delta)) - \lambda_2 \left(\frac{r}{2\|\xi_j\|_2} - K_{PLL} \right) - \lambda_3 (\rho_0 - \rho_0^{min})$$

At any KKT point (θ^*, λ^*) with $\lambda_i \geq 0$, the stationarity conditions are

1) $$\frac{\partial \mathcal{E}}{\partial \rho_0} = 2\|C_{reg}^{-1}\|_2^2 \rho_0 \left(C_n + \frac{C_1^2 \sigma_{EEG}^2}{\Delta^2} \right) - \lambda_3 = 0$$

2) $$\frac{\partial \mathcal{E}}{\partial \Delta} = -\frac{2\|C_{reg}^{(j)}\|_2^2 C_1^2 \rho_0^2 \sigma_{EEG}^2}{\Delta^3} + \lambda_1 \frac{\partial f}{\partial \Delta} = 0$$

3) $$\frac{\partial \mathcal{E}}{\partial K_{PLL}} = -\|C_{reg}^{(j)}\|_2^2 A_j^2 \sigma_\phi^2 \alpha \|\xi_j\|_2^2 \frac{2 - 2\alpha K_{PLL} \|\xi_j\|_2^2}{[\alpha K_{PLL} \|\xi_j\|_2^2 (2 - 2\alpha K_{PLL} \|\xi_j\|_2^2)]^2} + \lambda_2 = 0$$

Together with complementary slackness $\lambda_i (\text{constraint}) = 0$ for each $i = \{1, 2, 3\}$

Proof The objective $\mathcal{E}$ is C^∞ in θ on the feasible set; the constraints are smooth by assumptions. The KKT conditions follow by standard Lagrange multiplier theory. Note that ρ_0^{min} does not depend on ρ_0 itself (it is a function of r_0, δ, α only) so $\partial \rho_0^{min} / \partial \rho_0 = 0$ and simplifies as shown. The remaining conditions follow by differentiation of $\mathcal{E}(\theta)$.

Closed-Form Solutions

Proposition. Adopt the following tractable models consistent with assumptions.

$$f(r_0, \alpha) = \frac{c_f}{r_0^2} \qquad \rho_0^{min}(r_0, \alpha, \delta) = \frac{c_m}{\Delta}, \quad \Delta = r_0 - \delta$$

The model $\rho_0^{min} = c_m / \Delta$ captures the Neimark-Sacker activation condition: a larger spectral gap Δ requires less input drive to sustain oscillation. Under these, the ~~KKT~~ KKT system is solved by

$$K_{PLL}^* = \min\left(\frac{1}{\alpha \|\xi_j\|_2^2}, \frac{\Delta^*}{2\|\xi_j\|_2} \right)$$

$$\rho_0^* = \frac{c_m}{\Delta^*}$$

$\Delta^* = (r_0^* - \delta^*)$ where Δ^* solves the fixed point equation

$$\Delta^{*5} = \frac{2 c_1^2 \sigma_{EEG}^2 c_m^2}{c_n + \varepsilon_1^2 \sigma_{EEG}^2 \Delta^{*2}} \cdot \frac{r_0^{*3}}{2 c_f}$$

Proof. Step #1: Optimal K_{PLL}^* — $\frac{\partial \mathcal{E}}{\partial K_{PLL}}$ with $\lambda_2 \geq 0$ shows that $\partial \mathcal{E} / \partial K_{PLL} \leq 0$ at the optimum, meaning $\mathcal{E}_3$ is decreasing in K_{PLL} there. Since $\mathcal{E}_3$ is convex in K_{PLL} on $(0, 2/(\alpha \|\xi_j\|_2^2))$ with unconstrained minimum at $K_{PLL,unc}^* = 1/(\alpha \|\xi_j\|_2^2)$, the constrained optimum is

$$K_{PLL}^* = \min\left(K_{PLL,unc}^*, \frac{r^2}{2\|\xi_j\|_2} \right) = \min\left(\frac{1}{\alpha \|\xi_j\|_2^2}, \frac{\Delta^*}{2\|\xi_j\|_2} \right)$$

where in the second inequality we used the assumption $r^* \approx \Delta^*$.

Step #2: Optimal ~~$\Delta^* = $ Substitute~~ ρ_0^*. From $\partial \mathcal{E} / \partial \rho_0$ with

constraint C3 active ($\lambda_3 > 0$, $\rho_0 = \rho_0^{\min} = c_m/\Delta$):

$$2\|c_{reg}^{(j)}\|_2^2 \cdot \frac{c_m}{\Delta} \cdot \left(c_n + \frac{c_1^2 \sigma_{EEG}^2}{\Delta^2}\right) = \lambda_3 > 0$$

This confirms that the active constraint is consistence. Thus, $\rho_0^* = c_m/\Delta^*$

Step #3: Optimal Δ^* — Substitute $\rho_0 = c_m/\Delta$ into $\partial\mathcal{E}/\partial\Delta$ with $\lambda_1 > 0$ (constraint C1 active):-

$$\frac{-2\|c_{reg}^{(j)}\|_2^2 c_1^2 (c_m/\Delta)^2 \sigma_{EEG}}{\Delta^3} + \lambda_1 \frac{\partial f}{\partial \Delta} = 0$$

With $f(r_0, d) = c_f / r_0^2$ and $r_0 = \Delta + \delta^*$ (so $\partial f/\partial\Delta = -2c_f/r_0^3$), setting $\lambda_1 > 0$ and rearranging,

$$\frac{2\|c_{reg}^{(j)}\|_2^2 c_1^2 c_m^2 \sigma_{EEG}^2}{\Delta^5} = \frac{2\lambda_1 c_f}{r_0^3}$$

Eliminating λ_1 using $\partial\mathcal{E}/\partial\rho_0$ (which gives $\lambda_3 = 2\|c_{reg}^{(j)}\|_2^2 (c_m/\Delta)(c_n + c_1^2\sigma_{EEG}^2/\Delta^2)$) and matching scales yields the fixed-point equation for Δ^*. Since the right-hand side is decreasing in Δ^* and the left hand side is increasing, the fixed point equation has a unique solution, which can be found numerically by machine iteration or bisection on a compact interval $[\Delta_{min}, \Delta_{max}] = [\delta_{min}/2, 1 - \delta_{min}]$

The three error terms are not independently free. $\mathcal{E}_1$ and $\mathcal{E}_2$ are both proportional to ρ_0^2 so any reduction in one reduces the other. "The Pareto Front" between $\mathcal{E}_1 + \mathcal{E}_2$ and $\mathcal{E}_3$ is therefore 1-D, parametrized by K_{PID} above once $\rho_0 = \rho_0^{\min}$ and $\Delta = \Delta^*$ are fixed.

Summary of Results for Proportional Control

Theorem 3.10 (Joint Optimal Parameters) Under regularity conditions R1–R5 and the model $\rho_0^{\min} = c_m/\Delta$, the joint optimal parameter vector is

$$\boxed{\theta^* = \left(r_0^{max} - \epsilon_r,\ \delta_{min} + \epsilon_\delta,\ \min\left(\frac{1}{d\|e_{ij}\|_2^2},\ \frac{r_0^{max} - \delta_{min}}{2\|e_{ij}\|_2}\right),\ \frac{c_m}{r_0^{max} - \delta_{min}} + \epsilon_0\right)}$$

where $r_0^{max} = \sup\{r_0 : \rho_{ser} > f(r_0, d)\}$, δ_{min} is determined by the ESP constraint, and $\epsilon_r, \epsilon_\delta, \epsilon_0 > 0$.

The minimum acheivable tracking error is

$$\mathcal{E}(\theta^*) = \|C_{reg}^{(i)}\|_2^2 \left(\frac{C_m^2}{\Delta^{*2}} \left(C_n + \frac{C_i^2 \sigma_{\text{EECA}}^2}{\Delta^{*2}} + A_j^2 \, \mathbb{E}_\infty[\varepsilon_\phi^2]\big|_{K_{PLL}=K_{PLL}^*} \right) \right) \quad (11.1)$$

with $\Delta^* = r_o^{max} - \delta_{min}$ and $\mathbb{E}_\infty[\varepsilon_\phi^2]|_{K^*_{PLL}}$ from the steady-state mean squared phase error at $K_{PLL} = K_{PLL}^*$

Proof: The optimal r_o maximizes Δ and satisfies C1, giving $r_o^* = r_o^{max} - \epsilon_r$. With r_o^* fixed, δ^* should be minimized to maximize Δ, giving $\delta^* = \delta_{min} + \epsilon_\delta$ where δ_{min} is the smallest δ consistent with the ESP and the subdomain spectral radius requirement. The optimal $\rho_o^* = \rho_o^{max}(\Delta^*) + \epsilon_\rho$ follows from the achievable constraint C3 and properties, along with K_{PLL}^*. The minimum error is obtained by substituting θ^* into $\mathcal{E}(\theta)$'s error decomposition.

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